

Software Requirements Specifications

Project Title: Micro-Service Hub

Part 3: Functional Requirements

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3. Functional Requirements

The functional requirements define the functions and services that the **Micro-Service Hub Platform** shall provide to its users and administrators. The system shall support user registration, NID verification, service posting, service discovery, service application, price negotiation, real-time communication, online payment, service completion, reviews, dispute management, and administration.

3.1 User Registration and Login

FR-01: The system shall allow users to create an account by providing required information such as name, email/phone number, and password.

FR-02: The system shall allow registered users to log in using their valid credentials.

FR-03: The system shall validate user credentials before granting access to the system.

FR-04: The system shall provide password reset functionality for users who forget their passwords.

FR-05: The system shall allow users to update their profile information.

3.2 User Role Management

FR-06: The system shall support two primary user roles:

- Service Seeker
- Service Provider

FR-07: A registered user shall be able to use the platform as a service seeker or service provider.

FR-08: The system shall provide appropriate functions and permissions according to the user's role.

3.3 NID Verification

FR-09: The system shall allow users to submit the required NID information for identity verification.

FR-10: The system shall verify the submitted identity information through an available KYC/NID verification service or sandbox API.

FR-11: The system shall display the user's verification status as **Verified, Pending, or Rejected**.

FR-12: The system shall allow users to view their own verification status.

FR-13: The system shall use the verification status to establish trust between service seekers and service providers.

3.4 Service Posting

FR-14: The system shall allow service seekers to create and post micro-service requests.

FR-15: A service post shall contain information such as:

- Service title
- Service description
- Service category
- Location
- Required date/time
- Offered amount
- Additional instructions

FR-16: The system shall allow service seekers to edit their service posts before a provider is selected.

FR-17: The system shall allow service seekers to cancel their service posts when permitted by the service status.

FR-18: The system shall display active service posts to eligible service providers.

3.5 Service Discovery and Search

FR-19: The system shall allow service providers to browse available micro-services.

FR-20: The system shall allow users to search for services using keywords and categories.

FR-21: The system shall provide location-based service discovery to help users find nearby services.

FR-22: The system shall allow users to filter services based on criteria such as category, location, budget, and service status.

FR-23: The system shall display relevant service information before a user applies for a service.

3.6 Service Application

FR-24: The system shall allow service providers to apply for available service requests.

FR-25: A service provider shall be able to submit an application containing a message and relevant information.

FR-26: The service seeker shall be able to view applications received from service providers.

FR-27: The service seeker shall be able to select a suitable service provider from the received applications.

FR-28: The system shall notify the selected service provider when their application is accepted.

FR-29: The system shall update the service status after a provider is selected.

3.7 Price Offer and Negotiation

FR-30: The service seeker shall specify an initial amount they are willing to pay for the requested service.

FR-31: The service provider shall be able to view the amount offered by the service seeker.

FR-32: The service provider shall be able to accept the offered amount.

FR-33: The service provider shall be able to submit a counter offer with a different amount instead of accepting the initial offer.

FR-34: The service seeker shall be able to accept or reject the counter offer.

FR-35: The service seeker shall also be able to submit another amount in response to a counter offer.

FR-36: The system shall allow both parties to continue price negotiation until an amount is mutually agreed upon.

FR-37: The system shall maintain the history of offers and counter offers exchanged between the two parties.

FR-38: The system shall determine the final service amount only after both parties agree on an amount.

FR-39: The system shall notify users when an offer or counter offer is submitted, accepted, or rejected.

3.8 Real-Time Chat and Communication

FR-40: The system shall provide a real-time chat facility between the service seeker and service provider.

FR-41: The service seeker and service provider shall be able to exchange text messages in real time.

FR-42: Users shall be able to discuss service requirements, price, timing, location, and other relevant details through the chat system.

FR-43: The system shall maintain the chat history associated with a service.

FR-44: The system shall restrict service-related chat access to the relevant service seeker and service provider.

FR-45: The system shall notify users when a new chat message is received.

3.9 Location and Map Services

FR-46: The system shall allow users to provide a location related to a service.

FR-47: The system shall display service locations using an integrated map service such as Google Maps or OpenStreetMap.

FR-48: The system shall support location-based matching between service seekers and nearby service providers.

FR-49: The system shall allow users to view relevant location information before applying for or accepting a service.

3.10 Payment Processing

FR-50: The system shall allow the service seeker to make an online payment after the final service amount has been agreed upon.

FR-51: The system shall integrate with a supported payment gateway for processing online payments.

FR-52: The system shall hold the agreed payment amount securely until the service is completed and confirmed.

FR-53: The system shall release the held payment to the service provider after successful confirmation of the completed service.

FR-54: The system shall record payment status and transaction information.

FR-55: The system shall notify users about successful, failed, pending, or released payments.

FR-56: The system shall support sandbox/test payment APIs for academic development.

3.11 Service Completion and Confirmation

FR-57: The service provider shall be able to mark an assigned service as completed.

FR-58: The service seeker shall be able to view the service completion status.

FR-59: The service seeker shall be able to confirm whether the service has been completed successfully.

FR-60: The system shall release the held payment after successful confirmation.

FR-61: The system shall update the service status to **Completed** after confirmation.

FR-62: The system shall prevent the payment from being released before the required completion confirmation.

3.12 Review System

FR-63: The system shall allow the service seeker to write a review for the service provider after completion of a service.

FR-64: The system shall allow the service provider to write a review for the service seeker after completion of a service.

FR-65: The system shall store reviews associated with completed services.

FR-66: The system shall display relevant reviews on user profiles.

FR-67: The system shall not use a numerical or star-based rating system.

3.13 Dispute and Report Management

FR-68: The system shall allow users to report inappropriate behavior, fraudulent activity, or service-related problems.

FR-69: The system shall allow users to raise a dispute regarding a service or payment.

FR-70: The system shall allow users to provide relevant details when submitting a dispute or report.

FR-71: The system shall store submitted reports and dispute information.

FR-72: The administrator shall be able to review and manage disputes and reports.

FR-73: The administrator shall be able to take appropriate action regarding disputes according to system policies.

3.14 Notifications

FR-74: The system shall provide notifications for important service-related events.

FR-75: The system shall generate notifications for events such as:

- New service application
 - Application acceptance
 - New price offer
 - Counter offer
 - Offer acceptance or rejection
 - New chat message
 - Payment status update
 - Service completion request
 - Service confirmation
 - New review
 - Dispute/report status update
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3.15 Admin Dashboard

FR-76: The system shall provide a secure dashboard for administrators.

FR-77: The administrator shall be able to view and manage registered users.

FR-78: The administrator shall be able to monitor service posts and service activities.

FR-79: The administrator shall be able to monitor user verification statuses.

FR-80: The administrator shall be able to monitor payment and transaction information.

FR-81: The administrator shall be able to review reports, disputes, and complaints.

FR-82: The administrator shall be able to suspend or restrict accounts that violate system rules.

3.16 Service Status Management

FR-83: The system shall maintain the status of each service throughout its lifecycle.

FR-84: The service status shall include states such as:

- Posted
- Application Received
- Provider Selected
- Offer Submitted
- Counter Offer Submitted
- Price Agreed
- Payment Pending

- Payment Held
- In Progress
- Completed
- Confirmed
- Cancelled
- Disputed

FR-85: The system shall automatically update the service status based on relevant user actions.

3.17 Service History

FR-86: The system shall maintain a history of services posted, applied for, accepted, and completed by each user.

FR-87: Users shall be able to view their previous service activities.

FR-88: Users shall be able to view relevant offers, counter-offer, chat, payment, and review history associated with their services.

3.18 Overall System Workflow

FR-89: The system shall support the complete service workflow:

Register → Login → NID Verification → Post/Find Service → Apply → Initial Amount Offer → Accept/Counter Offer → Real-Time Chat → Price Agreement → Payment → Service Completion → Confirmation → Payment Release → Review

FR-90: The system shall ensure that the final payment amount is determined only after the service seeker and service provider mutually agree on an amount.

FR-91: The system shall ensure that the payment is processed according to the mutually agreed final amount.

FR-92: The system shall maintain the necessary records of the complete service transaction from service posting to service completion and review.